

$$\begin{aligned} & \left(-\frac{1}{54} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_d^{\mathbf{p}}] + \frac{5}{144} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_d^{\mathbf{p}}] - \frac{2}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_d^{\mathbf{p}}] - \frac{1}{18} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_e^{\mathbf{p}}] + \right. \\ & \frac{5}{48} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_e^{\mathbf{p}}] - \frac{2}{45} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_e^{\mathbf{p}}] - \frac{1}{144} \sum_{\mathbf{p}} (4 g_1^4 + 9 g_2^4) \text{LF}_{3,0} [m_l^{\mathbf{p}}] + \\ & \frac{1}{96} \sum_{\mathbf{p}} (5 g_1^4 + 6 g_2^4) \text{LF}_{4,-1} [m_l^{\mathbf{p}}] - \frac{1}{45} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_l^{\mathbf{p}}] + \\ & \left. \left(\frac{3}{4} g_2^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} - \frac{1}{432} \sum_{\mathbf{p}} (4 g_1^4 + 81 g_2^4) \right) \text{LF}_{3,0} [m_q^{\mathbf{p}}] + \right. \\ & \left(-\frac{3}{4} g_2^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} + \frac{1}{288} \sum_{\mathbf{p}} (5 g_1^4 + 54 g_2^4) \right) \text{LF}_{4,-1} [m_q^{\mathbf{p}}] - \frac{1}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_q^{\mathbf{p}}] - \\ & \frac{2}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_u^{\mathbf{p}}] + \frac{5}{36} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_u^{\mathbf{p}}] - \frac{8}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_u^{\mathbf{p}}] - \frac{1}{36} g_1^4 \text{LF}_{3,0} [\tilde{\mu}] - \\ & \frac{1}{24} g_1^4 \text{LF}_{4,-1} [\tilde{\mu}] + \frac{2}{45} g_1^4 \text{LF}_{5,-2} [\tilde{\mu}] - \frac{1}{32} g_1^4 \text{LF}_{2,2,-1} [m_1, \tilde{\mu}] - \frac{1}{16} g_1^4 m_1^2 \text{LF}_{2,2,0} [m_1, \tilde{\mu}] - \\ & \frac{17}{32} g_2^4 \text{LF}_{2,2,-1} [m_2, \tilde{\mu}] - \frac{1}{16} g_2^4 m_2^2 \text{LF}_{2,2,0} [m_2, \tilde{\mu}] + g_2^4 \text{LF}_{3,2,-2} [m_2, \tilde{\mu}] - \\ & \frac{3}{4} g_2^4 m_2^2 \text{LF}_{3,2,-1} [m_2, \tilde{\mu}] - \frac{1}{2} g_2^4 \text{LF}_{3,3,-3} [m_2, \tilde{\mu}] + \frac{1}{2} g_2^4 m_2^2 \text{LF}_{3,3,-2} [m_2, \tilde{\mu}] - \\ & \frac{1}{2} g_2^4 \text{LF}_{4,2,-3} [m_2, \tilde{\mu}] + \frac{1}{2} g_2^4 m_2^2 \text{LF}_{4,2,-2} [m_2, \tilde{\mu}] - \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{2,2,0} [m_d^{\mathbf{r}}, m_q^{\mathbf{p}}] + \\ & \frac{1}{12} g_1^2 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{3,2,-1} [m_d^{\mathbf{r}}, m_q^{\mathbf{p}}] - \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{2,2,0} [m_e^{\mathbf{r}}, m_l^{\mathbf{p}}] + \\ & \frac{1}{12} g_1^2 \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{3,2,-1} [m_e^{\mathbf{r}}, m_l^{\mathbf{p}}] + \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{3,1,0} [m_l^{\mathbf{p}}, m_e^{\mathbf{r}}] + \\ & \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{3,2,-1} [m_l^{\mathbf{p}}, m_e^{\mathbf{r}}] - \frac{1}{8} \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} (3 g_1^2 + g_2^2) \text{LF}_{4,1,-1} [m_l^{\mathbf{p}}, m_e^{\mathbf{r}}] + \\ & \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{5,1,-2} [m_l^{\mathbf{p}}, m_e^{\mathbf{r}}] + \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{3,1,0} [m_q^{\mathbf{p}}, m_d^{\mathbf{r}}] + \\ & \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{3,2,-1} [m_q^{\mathbf{p}}, m_d^{\mathbf{r}}] - \frac{1}{8} \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} (5 g_1^2 + 3 g_2^2) \text{LF}_{4,1,-1} [m_q^{\mathbf{p}}, m_d^{\mathbf{r}}] + \\ & \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{5,1,-2} [m_q^{\mathbf{p}}, m_d^{\mathbf{r}}] - \frac{3}{4} \overline{y_u^{\text{pr}}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}} \text{LF}_{2,1,0} [m_q^{\mathbf{p}}, m_q^{\mathbf{s}}] + \\ & \frac{3}{4} \overline{y_u^{\text{pr}}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}} \text{LF}_{3,1,-1} [m_q^{\mathbf{p}}, m_q^{\mathbf{s}}] + \frac{1}{12} g_1^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{2,2,0} [m_q^{\mathbf{p}}, m_u^{\mathbf{r}}] + \\ & \frac{3}{4} g_2^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{3,1,0} [m_q^{\mathbf{p}}, m_u^{\mathbf{r}}] - \frac{1}{24} \overline{a_u^{\text{pr}}} a_u^{\text{pr}} (g_1^2 + 9 g_2^2) \text{LF}_{3,2,-1} [m_q^{\mathbf{p}}, m_u^{\mathbf{r}}] - \\ & \frac{3}{4} g_2^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{4,1,-1} [m_q^{\mathbf{p}}, m_u^{\mathbf{r}}] - \frac{1}{12} g_1^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{3,1,0} [m_u^{\mathbf{r}}, m_q^{\mathbf{p}}] - \\ & \frac{1}{12} g_1^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{3,2,-1} [m_u^{\mathbf{r}}, m_q^{\mathbf{p}}] - \frac{1}{4} g_1^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{4,1,-1} [m_u^{\mathbf{r}}, m_q^{\mathbf{p}}] + \\ & \frac{1}{2} g_1^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{5,1,-2} [m_u^{\mathbf{r}}, m_q^{\mathbf{p}}] - \frac{1}{16} g_1^4 \text{LF}_{3,2,-2} [\tilde{\mu}, m_1] + \frac{1}{16} g_1^4 m_1^2 \text{LF}_{3,2,-1} [\tilde{\mu}, m_1] + \\ & \frac{5}{24} g_1^4 \text{LF}_{4,1,-2} [\tilde{\mu}, m_1] + \frac{1}{16} g_1^4 \text{LF}_{4,2,-3} [\tilde{\mu}, m_1] - \frac{1}{16} g_1^4 m_1^2 \text{LF}_{4,2,-2} [\tilde{\mu}, m_1] - \\ & \frac{1}{6} g_1^4 \text{LF}_{5,1,-3} [\tilde{\mu}, m_1] + \frac{3}{16} g_2^4 \text{LF}_{3,2,-2} [\tilde{\mu}, m_2] + \frac{1}{16} g_2^4 m_2^2 \text{LF}_{3,2,-1} [\tilde{\mu}, m_2] + \\ & \frac{5}{8} g_1^2 g_2^2 \text{LF}_{4,1,-2} [\tilde{\mu}, m_2] + \frac{1}{16} g_2^4 \text{LF}_{4,2,-3} [\tilde{\mu}, m_2] - \frac{1}{16} g_2^4 m_2^2 \text{LF}_{4,2,-2} [\tilde{\mu}, m_2] - \\ & \frac{1}{2} g_1^2 g_2^2 \text{LF}_{5,1,-3} [\tilde{\mu}, m_2] + g_1^2 g_2^2 \text{LF}_{2,2,1,-2} [m_2, \tilde{\mu}, m_1] + \frac{3}{4} m_1 m_2 g_1^2 g_2^2 \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_1] - \\ & \frac{1}{2} g_1^2 g_2^2 \text{LF}_{3,2,1,-3} [m_2, \tilde{\mu}, m_1] - \frac{1}{2} m_1 m_2 g_1^2 g_2^2 \text{LF}_{3,2,1,-2} [m_2, \tilde{\mu}, m_1] + \\ & \frac{3}{4} \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{2,2,1,-1} [m_q^{\mathbf{p}}, m_q^{\mathbf{s}}, m_d^{\mathbf{r}}] - \frac{3}{4} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \overline{a_u^{\text{pr}}} a_u^{\text{sr}} \text{LF}_{2,1,1,0} [m_q^{\mathbf{p}}, m_q^{\mathbf{s}}, m_u^{\mathbf{r}}] + \\ & \frac{3}{4} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \overline{a_u^{\text{pr}}} a_u^{\text{sr}} \text{LF}_{3,1,1,-1} [m_q^{\mathbf{p}}, m_q^{\mathbf{s}}, m_u^{\mathbf{r}}] + \frac{3}{8} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \overline{a_u^{\text{pr}}} a_u^{\text{sr}} \text{LF}_{2,2,1,-1} [m_q^{\mathbf{p}}, m_u^{\mathbf{r}}, m_q^{\mathbf{s}}] - \\ & \frac{3}{4} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \overline{a_u^{\text{pr}}} a_u^{\text{sr}} \text{LF}_{2,1,1,0} [m_q^{\mathbf{s}}, m_q^{\mathbf{p}}, m_u^{\mathbf{r}}] + \frac{3}{4} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \overline{a_u^{\text{pr}}} a_u^{\text{sr}} \text{LF}_{3,1,1,-1} [m_q^$$